Fiscal Year:	FY 2020	**Task Last Updated:**	FY 04/03/2020
PI Name:	Allaway, Heather Ph.D.		
Project Title:	Will Use of Long-Acting, Reversible Contraceptives Mitigate Bone Loss in Female Rats During Simulated Exploration Class Missions?		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	70803-0001	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 TRI-RFA-17-01: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	01/01/2018	**End Date:**	12/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2020 per March 2020 report from TRISH (Ed., 4/7/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bloomfield, Susan Ph.D. (MENTOR: Texas A&M University)		
Grant/Contract No.:	NNX16AO69A-P0204		
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Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Astronauts who spend a long time in weightlessness typically lose bone and muscle mass -- which may be very dangerous for astronauts arriving on Mars. Hormonal contraceptive (HC) methods routinely used female astronauts, and women worldwide, to suppress ovarian function and menstrual cycling. Use of HC decreases natural estrogen and progesterone exposure and the positive influence of both hormones on bone health. There is a critical gap in knowledge surrounding long-acting, reversible contraceptive (LARC; progestin-only) use, an HC option chosen with increasing frequency, and bone geometry and cellular activity. There is no knowledge about the interaction of LARC use and microgravity on bone health. Specific Aim 1: Determine the dose dependent impact of an etonogestrel (ENG) LARC implant on ovarian function in intact, young adult female rats over 28 days of exposure. Specific Aim 2: Determine the combined effects of an ENG LARC implant and skeletal disuse (hindlimb unloading; HU) on bone health in intact, young adult, female rats.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will have impact with the shift from short to long duration and exploration class missions and has the potential to shift clinical practice. In female astronauts of reproductive age, the impact of spaceflight on gonadal function is complicated by the use of hormonal contraception to suppress the menstrual cycle pre-flight and during flight. Ground based studies have failed to directly account for the impact of long term use of hormonal contraception on bone health. We want to facilitate the improvement of best practice by working to ensure that long term use of hormonal contraception methods does not impair health. This study will be the first to explore the impact of LARC use on bone health using an animal model of microgravity exposure and will use measures of bone geometry, strength, and cellular activity. The results of this study will be translatable to the astronaut corps as well as millions of women around the world who use hormonal contraception. Additionally, this project is the development of a novel model to assess the impact of hormonal contraceptive use on bone health and future fracture risk.
Task Progress:	[Ed. note April 2020: Report submitted by TRISH to Task Book in March 2020; covers reporting as of November 2019.] Astronauts who spend a long time in weightlessness typically lose bone and muscle mass -- which may be very dangerous for astronauts arriving on Mars. Hormonal contraceptive (HC) methods are routinely used by female astronauts, and women worldwide, to suppress ovarian function and menstrual cycling. Use of HC decreases natural estrogen and progesterone exposure and the positive influence of both hormones on bone health. There is a critical gap in knowledge surrounding long-acting, reversible contraceptive (LARC; progestin-only) use, an HC option chosen with increasing frequency, and bone geometry and cellular activity. There is no knowledge about the interaction of LARC use and microgravity on bone health. SA1: Determine the dose dependent impact of an etonogestrel (ENG) LARC implant on ovarian function in intact, young adult female rats over 28 days of exposure. We hypothesize that: H1A: ENG LARC implantation will initially suppress endogenous progesterone and estradiol in a dose dependent manner; H1B: ENG LARC implantation will continue to suppress endogenous progesterone while estradiol will rise to moderate cycle concentration after 14 days and remain elevated thereafter in a dose dependent manner; H1C: ENG LARC implantation will reduce ovarian volume and suppress ovulation (i.e., increased number of small ovarian follicles and no corpora lutea), as assessed by ultrasound biomicroscopy, in a dose dependent manner; H1D (exploratory): ENG LARC implantation will decrease serum measures of osteoclast activity (CTx) and increase osteoblast activity (P1NP) in a dose dependent manner. SA2: Determine the combined effects of an ENG LARC implant and skeletal disuse (hindlimb unloading; HU) on bone health in intact, young adult, female rats. We hypothesize that: H2A: ENG LARC implantation will blunt the decreases in bone mineral density (BMD), cortical thickness, cancellous microarchitecture, and mechanical strength observed with 6 weeks of HU; H2B: ENG LARC implantation will blunt the decreases in bone formation rate and osteoid surface and blunt the increases in osteoclast surface observed with 6 weeks of HU; H2C (exploratory): ENG LARC implantation will blunt the reduction in estrogen and progesterone receptor density in osetoblasts and osteocytes with 6 weeks of HU. Animal husbandry for SA1 was completed 06/01/2018. Animals arrived at 16 weeks of age, had the ENG LARC/placebo implant inserted at 20 weeks of age, and were sacrificed at 24 weeks of age. The ENG LARC implants released doses of ENG at rates of 0.00 ug/d (placebo), 0.09 ug/d (low), 0.17 ug/d (medium), or 0.30 ug/d (high). Animal body weight and food consumption did not differ among treatment groups ($p>0.150$). Uterine and ovarian weights at sacrifice (a biomarker of estradiol exposure) did not differ among treatment groups ($p>0.150$). Ovarian volume and number of ovulatory sized follicles (>400 um) did not differ among treatment groups across the study ($p>0.071$; H1C). The ultrasound biomicroscope images were not clear in all animals to monitor the smallest follicle sizes we intended to assess. Serum estradiol concentrations did not differ among treatment groups across the study ($p=0.216$; H1A/B). Evaluated as total number of days or percent total time, the time spent in proestrus, estrus, metestrus, and diestrus did not differ among the treatment groups prior to implant insertion ($p>0.30$). Following implant insertion, time spent in estrus was reduced and diestrus increased for the medium and high dose treatment groups ($p<0.001$), with no differences in time spent in proestrus or metestrus ($p>0.08$). In the first 2 weeks following implant insertion there is an apparent dose dependent shift toward irregularity and acyclicity ($p<0.001$), which continued into the last 2 weeks of implant insertion ($p=0.02$). The shift in cycle classifications was similar to the shifts observed in women over the 3 years of implant use. Based on the more controlled suppression of estrus cyclicity, the high dose of ENG LARC was chosen as the dose to be used in SA2. Animal husbandry for SA2 was completed 12/06/2018. Animals arrived at 16-18 weeks of age, had the ENG LARC/placebo implant inserted at 20-22 weeks of age. The weight bearing/HU phase of the study was initiated at 21-23 weeks of age and sacrifice occurred at 27-29 weeks of age. Animal body weight was reduced over time in the HU animals starting week 3 of HU ($p<0.001$). Average daily food consumption was increased in the HU animals starting week 2 of HU ($p<0.001$). There were no differences among groups in ovarian weights taken at sacrifice ($p>0.30$) and HU animals had smaller solei than WB animals ($p<0.001$). LARC implantation did not prevent nor worsen changes in volumetric BMD or cortical thickness of the tibia from pre-HU to post-HU assessed by in vivo pQCT (H2A). Tibial mechanical strength was partially protected by LARC implantation ($p=0.012$; H2A). HU animals had reduced endosteal and periosteal bone formation rate per unit of bone surface (BFR/BS) at the tibia diaphysis ($p<0.02$; H2B). At the tibia metaphysis, animals with the LARC implant had more osteoid compared with placebo animals ($p=0.04$; H2B). There were no significant effects of LARC or HU on cancellous BFR/BS or osteoclast surface (Oc.S)/BS ($p>0.30$; H2B) at the tibia metaphysis.

	In the coming year we will finish SA2 analyses for serum CTx and P1NP, IHC for ER and PR (H2C), and mechanical testing on the femoral neck. Once tissue analysis is complete we will submit SA2 for publication. The paper for SA1 is in preparation for submission. Key findings thus far have not led to the revision of the original aims and hypotheses; however, cost and availability of assays has prevented the completion of the progesterone, CTx, and P1NP components of SA1. For the next year we have added 2 additional aims. SA3: Determine the combined effects of LARC and overloading on the health of the humeri of female rats in HU. SA4: Determine the combined effect of a LARC implant and HU on the gastrocnemius muscle of female rats.
Bibliography Type:	Description: (Last Updated: 01/28/2026)
Abstracts for Journals and Proceedings	Allaway H, Pierson R, Invik J, Bloomfield S. "Development of an animal model of long-acting, reversible contraception at physiologic equivalent doses." ENDO 2019, Endocrine Society Annual Meeting, New Orleans, Louisiana, March 23-26, 2019. Journal of the Endocrine Society. 2019 Apr-May;3(S1):SAT-LB039. https://doi.org/10.1210/js.2019-SAT-LB039 , Apr-2019
Abstracts for Journals and Proceedings	Allaway HCM, Little SE, Hogan HA, Bloomfield SA. "Impact of long-acting, reversible contraceptives on bone density during simulated microgravity." American College of Sports Medicine 66th Annual Meeting, Orlando, FL May 28-June 1, 2019. Medicine & Science in Sports & Exercise. 2019 Jun;51(6 Suppl):405-6 (abstract 1485). https://doi.org/10.1249/01.mss.0000561716.33913.d1 , Jun-2019
Abstracts for Journals and Proceedings	Vargas XS, Allaway HCM, Little SE, Bloomfield S. "Do implantable, progestin only contraceptives alter the bone response to unloading?" Texas Chapter of the American College of Sports Medicine 2019 Annual Meeting, Fort Worth, Texas, February 28-March 1, 2019. International Journal of Exercise Science: Conference Proceedings. 2019;2(11):122. Available at: https://digitalcommons.wku.edu/ijesab/vol2/iss11/122 ; accessed 4/7/2020. , Feb-2019
Awards	Allaway H. "Alice L. Jee Young Investigator Award, July 2019." Jul-2019
Awards	Allaway H. "Postdoctoral Scholar Travel Award, August 2019." Aug-2019